NEWTON SCHOOL OF TECHNOLOGY

SHADOW INSIGHT X

NYC Real Estate Sales Analysis:

Capstone Project - 2

TEAM - MEMBERS

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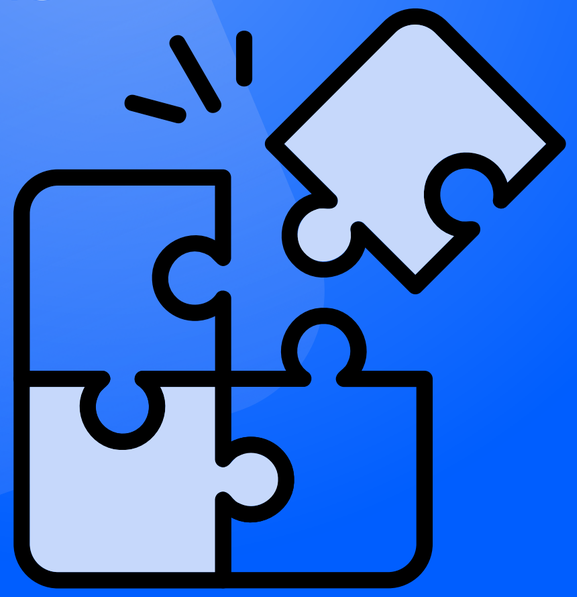
PROBLEM STATEMENT

- NYC real estate market is highly complex & non-transparent
- Prices vary across:
- Boroughs
- Property types
- Time (seasonality)
- Stakeholders rely on averages → misleading decisions

Goal:

Identify key factors driving property prices



PROJECT OBJECTIVE

- Identify key value drivers (location, size, type)
- Segment market (borough, property type, price range)
- Build data-driven insights & baseline model

- Support:

  Investors

  Developers

  Policymakers

DATASET OVERVIEW

- Source: NYC Rolling Sales Dataset
- ~84,000 transactions
- 21 columns

Key Features:

- Sale Price
- Borough
- Square Feet
- Units (Residential/Commercial)
- Sale Date

DATA CLEANING & ETL

- Removed:
- $0 & invalid sales
- Outliers (1st–99th percentile)
- Converted data types (string → numeric)
- Handled missing values

- **Feature Engineering:**

1. Price per sqft
2. Building age
3. Floor Area Ratio (FAR)
4. Price brackets

KPI FRAMEWORK

- **Valuation Metrics**
  - Median Price: $636K
  - Price per sqft: $350
- **Market Metrics**
  - Monthly sales volume
  - Borough market share
- **Property Metrics**
  - Building age
  - FAR (density indicator)

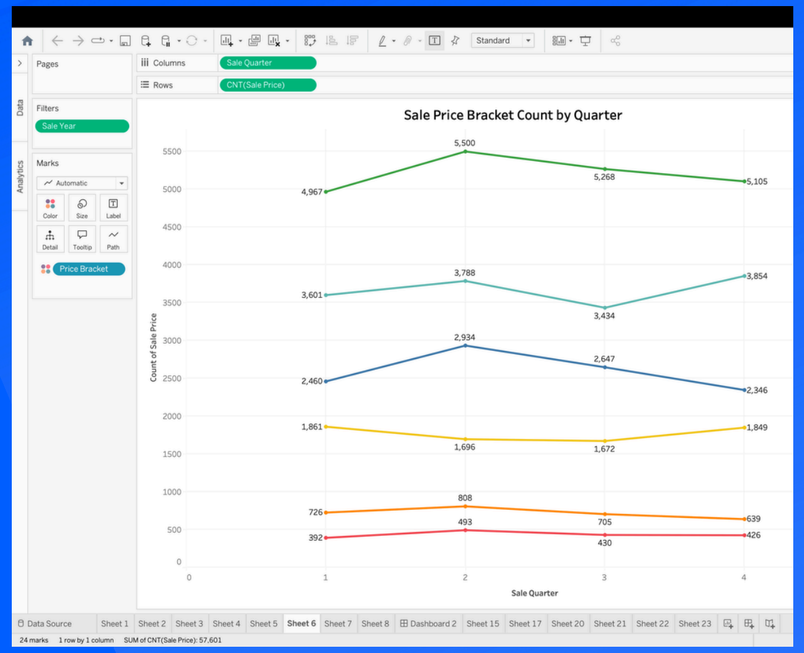
KEY EDA INSIGHTS

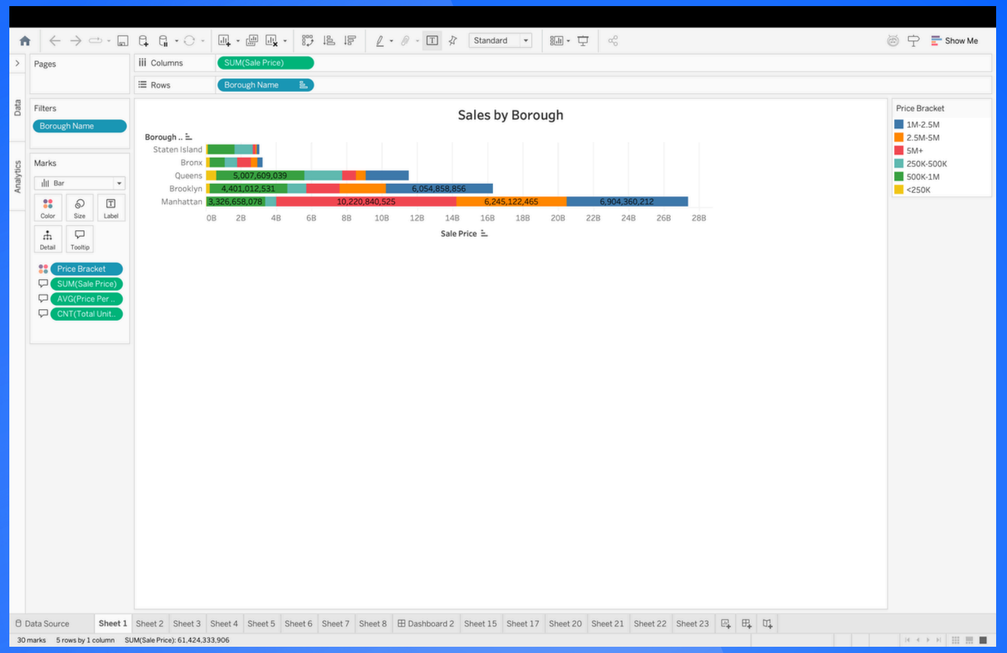
- Mean price ($1.07M) > Median ($636K) → skewed market
- Manhattan price per sqft = 3× Bronx
- 62% transactions in $250K–$1M range
- Summer ↑ sales volume, NOT price

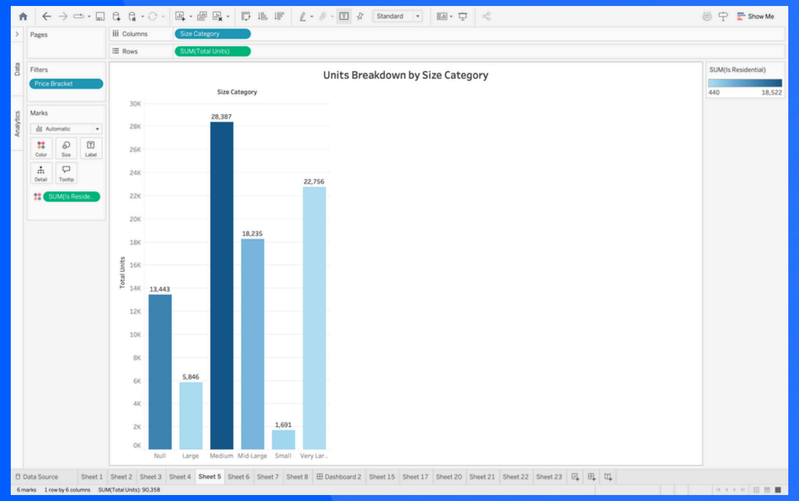
STATISTICAL ANALYSIS

- ANOVA Test:
  → Borough significantly affects price
- T-Test:
  → Residential ≠ Commercial
- Chi-Square:
  → Building class impacts price bracket

VISUAL INSIGHTS







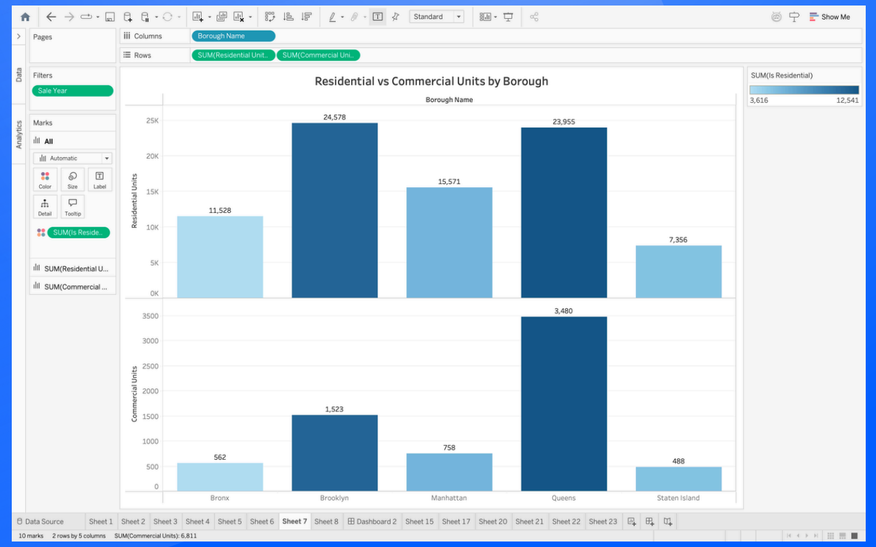
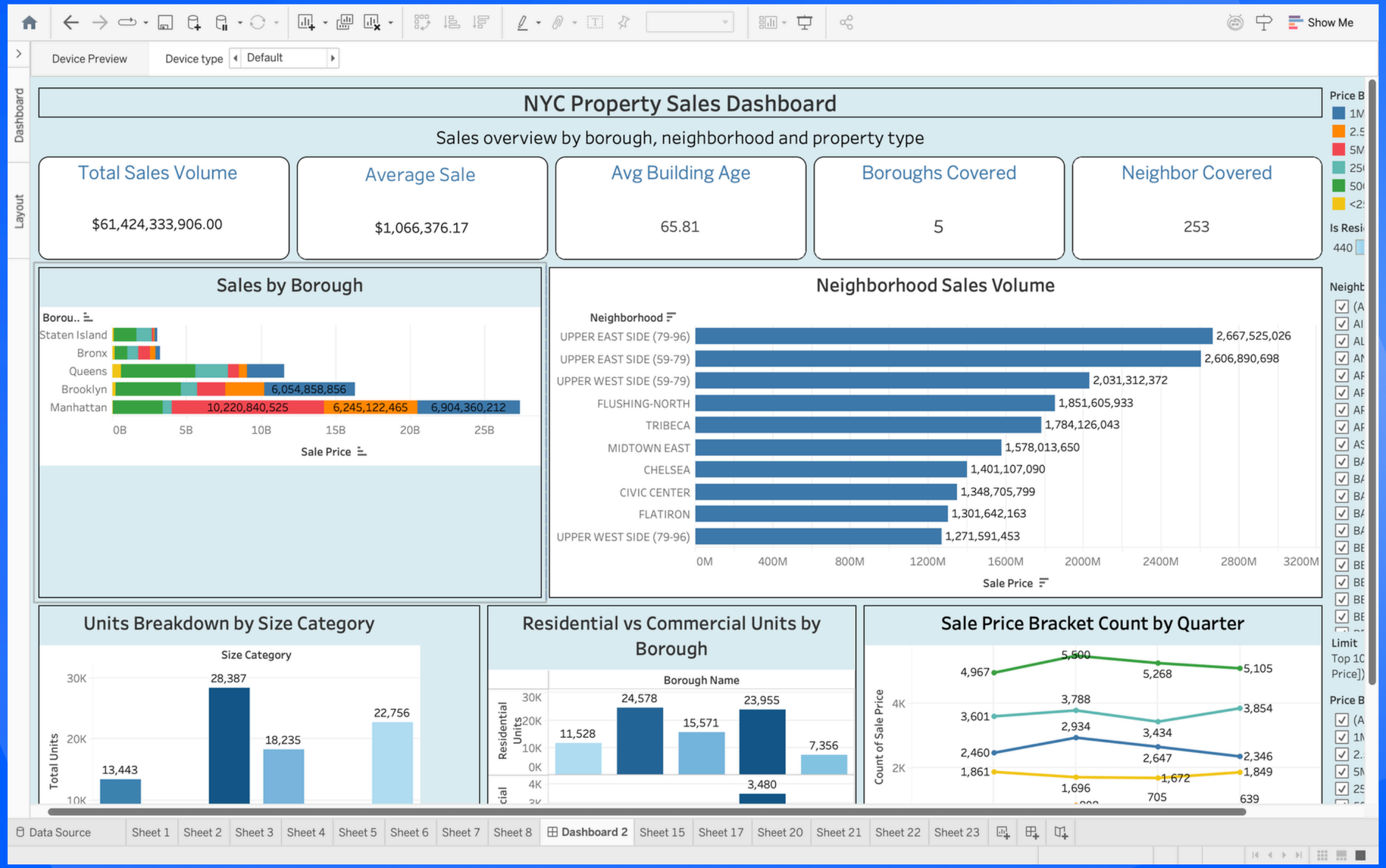


TABLEAU DASHBOARD



BUSINESS RECOMMENDATIONS

1. Use borough-specific pricing models
2. Sell in April–July (high demand)
3. Invest in pre-war building renovations
4. Separate commercial & residential analysis
5. Focus on high FAR properties

CONCLUSION & IMPACT

1. **NYC is not one market → it's 5 markets**
2. **Location is the biggest driver**
3. **Data-driven decisions > assumptions**

Impact :

1. **Better investment strategies**
2. **Smarter development planning**
3. **Improved valuation accuracy**

THANK YOU

TEAM - SHADOW INSIGHT X